O-I-520B
August 16, 1977
SUPERSEDING
O-I-520a
February 27, 1967

FEDERAL SPECIFICATION

INSECTICIDE, DIAZINON, EMULSIFIABLE CONCENTRATE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a diazinon emulsifiable concentrate which, when diluted with a specified amount of water, is suitable for the residual control of insects resistant to chlorinated hydrocarbon insecticides.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon)

Federal Standards:

Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies)
Fed. Test Method Std. No. 791	-	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

FSC 6840

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are availabl without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS:

Federal Insecticide, Fungicide, and Rodenticide Act

(Title 7 United States Code Section 135 et seq.) as amended by the Federal Environmental Pesticide Control Act of 1972 (PL 92-516) and any regulations promulgated thereunder by the administrator of the Environmental Protection Agency.)

(Application for copies should be addressed to the Environmental Protection Agency, Pesticide Registration Divison, 401 M Street, S.W., Washington, DC 20460.)

Title 49, Code of Federal Regulations (CFR) Parts 170 through 179, Hazardous Materials Regulations.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 170 to 179".)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Dept., 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition. The emulsifiable concentrate shall be composed of stabilized diazinon, a solvent and a nonionic emulsifier and shall contain not less than 47.5 percent by weight of pure diazinon (0,0-diethyl 0-(2-isopropyl-6-methyl-4-pyrimidinyl) phosphorothioate).

3.1.1 Diazinon content. The technical grade diazinon used in formulating this insecticide shall contain approximately 90 percent diazinon.

3.1.2 Solvent. The solvent used shall be an aromatic petroleum derivative which shall enable the finished formulation to meet the physical and chemical requirements of this specification.

3.1.3 Nonionic emulsifying agent. The nonionic emulsifier shall be completely miscible in the finished formulation, shall have no adverse effects on the diazinon ingredient and shall enable the formulation, when diluted with water, to meet the specified emulsion stability requirement (see 3.2).

3.2 End item.

3.2.1 Appearance. The insecticide shall (1) at 80°F (26.7°C) be clear and homogeneous with no sediment, and (2) at 32°F (0°C) show no more than slight deposition and be readily soluble on warming to room temperature (73° ± 5°F/ 22.8 ± 2.8°C) when tested as specified in 4.3.4.

3.2.2 Emulsion stability. The emulsion formed upon dilution of the formulation with hard water shall show not more than 1 ml separation per 100 ml of solution when tested as specified in 4.3.4.

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3.2.3 Diazinon content. The insecticide shall contain a minimum of 47.5 percent diazinon when tested as specified in 4.3.4.

3.2.4 Staining. The insecticide shall show no staining when tested as specified in 4.3.4.

3.2.5 Flashpoint. The insecticide shall have a minimum flashpoint of 100°F (37.8°C) when tested as specified in 4.3.4.

3.2.6 Residual odor. The insecticide shall have a mild and inoffensive odor when tested as specified in 4.3.4. The use of odor depressing or neutralizing agents to mask the residual odor requirement is not permitted.

3.2.7 pH. The pH of the insecticide shall have a minimum value of 6.0 and a maximum value of 9.0 when tested as specified in 4.3.4.

3.2.8 Water content. The insecticide concentrate shall have a maximum of 0.1 ml per 100 ml of concentrate when tested as specified in 4.3.4.

3.2.9 Compatability with kerosene and diesel oil. When mixed with kerosene or diesel oil, the insecticide shall show not more than a slight amount of sediment when tested as specified in 4.3.4.

3.3 Labeling. The labeling shall be registered with the Pesticide Registration Division, Environmental Protection Agency, and shall be in compliance with the requirements of the Federal Insecticide Fungicide and Rodenticide Act, as amended, of 1972.

3.3.1 Label data. Label data shall be durably and legibly marked on the main body, by lithographing or stenciling with non-corrosive, weather-resistant, waterproof ink which will not smear or rub off during handling. Label data shall include, but not be limited to the following:

(a) The name and percent by weight of each active ingredient.
(b) The total percent by weight of inactive ingredients.
(c) National stock number, Federal specification number, contract number, and lot number.
(d) Manufacturer's name and address.
(e) EPA registration and establishment number.
(f) Directions for use: As a minimum, labeling instructions must include suitable directions for: Household Insect Control (cockroaches, ants, silverfish, carpet beetles, spiders, scorpions, sawtooth grain beetles, flour beetles, rice weevils, cigarette beetles, drugstore beetles, Indian meal moths); Control of Brown Dog Ticks (in buildings) Insect Control Within Food and Non-Food Areas of Food Handling Establishments; Ornamental Insect Control; Lawn Insect Control (ants, brown dog ticks, clover mites, crickets, earwigs, fleas, millipedes, sawbugs, springtails).

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(g) **Precautionary information:** WARNING. KEEP OUT OF REACH OF CHILDREN. To be applied only under supervision of trained personnel. May be fatal if swallowed. May be absorbed through the skin. Do not breathe spray mist. Causes eye and skin irritation. Do not get in eyes, on skin, or on clothing. Wash thoroughly with soap and water after handling. Avoid contamination of feed or foodstuffs. DO NOT USE OR STORE NEAR HEAT OR OPEN FLAME.

CONTAINER DISPOSAL

1. Empty container in normal manner and allow to drain for one minute into the spray or mix tank.

2. Add one quart (946 ml) of rinse solution (water or designated spray carrier) per gallon (3.8 l) container size.

3. Replace closure.

4. Rotate and upend container to get rinse over all interior surfaces.

5. Drain rinse into the spray or mix tank.

6. Repeat steps 1 through 5 above for a total of 3 rinses. After the third rinse allow the reused container to drain for one minute into the spray or mix tank.

7. Crush the rinsed container and bury in a sanitary landfill in conformance with Federal, State and Local Standards, or recycle through a properly equipped metal reclaiming firm.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the cited requirements of the referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.3. The lot size for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled containers for the examinations in 4.3.1.1.

4.3.1.1 Examination of filled container for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine that the preservation-packaging, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container, fully prepared for delivery with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Marking and labeling	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flap, loose strapping, inadequate stapling, improper taping. Bulged or distorted container.
Contents	Number of pails per shipping container is less than specified.

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4.3.3 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination	Inspection level	AQL
4.3.1.1	S-2	N/A
4.3.2	S-2	2.5

4.3.4 Testing of the end item. The end item shall be tested for the characteristics specified in table I. The lot size shall be expressed in gallons of insecticide. The sample unit for testing shall be a 1 gallon composite obtained by combining and thoroughly mixing equal portions from containers selected at random throughout the lot. The composite shall be placed in a clean, dry, sealed container. Care shall be exercised to prevent contamination or alteration of the insecticide during the sampling, compositing, storage and testing. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE I. Instructions for testing of the end item

Characteristic	Requirement	Test method	Pass or fail	Results reported numerically to nearest
Appearance	3.2.1	4.4.1, 4.4.2	X	
Emulsion stability	3.2.2	4.4.5		1 ml
Diazinon content	3.2.3	4.4.3		0.1 percent
Staining	3.2.4	4.4.6	X	
Flashpoint	3.2.5	4.4.4		1°F (0.6°C)
Residual odor	3.2.6	4.4.6	X	
pH	3.2.7	4.4.7		0.1 pH unit
Water content	3.2.8	4.4.8		0.1 ml
Compatibility with kerosene and diesel oil	3.2.9	4.4.9	X	

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4.4 Test methods.

4.4.1 Appearance at 80°F (26.7°C). With the sample at 80° ± 1°F (26.7 ± 0.6 ..., the insecticide shall be examined visually by transmitted light for clarity, homogeneity, and sediment.

4.4.2 Appearance at 32°F (0°C). The appearance of the insecticide at 32°F (0°C) may be determined by transferring 50 ml of the insecticide to a 50 ml glass-stoppered cylinder, storing for 1 week at 32° ± 1°F (0° ± 0.6°C). The sample shall then be examined by transmitted light for clarity, phase formation, sediment, and deposition.

4.4.3 Diazinon content. The diazinon content shall be determined as follows:

4.4.3.1 Reagents. The following reagents shall be used:

Petroleum ether: 30° to 60°C boiling range.

Wash solution - To 500 ml of distilled water add 500 ml of a saturated and filtered sodium chloride solution and 6.5 ml of 0.1N sodium hydroxide.

Acetic Anhydride - Reagent grade.

Nitromethane - Redistilled commercial grade.

Acetic Acid - Glacial, reagent grade.

Perchloric acid (0.1N) - Combine 9.0 ml of perchloric acid (70 to 72 percent reagent grade) with 100 ml of glacial acetic acid in a 1-liter volumetric flask. Mix well, add 20 ml of acetic anhydride, and make to the mark with additional glacial acetic acid. Allow to stand at least 24 hours and standardize against potassium acid phthalate. 1/

Potassium acid phthalate - US National Bureau of Standards primary standard grade.

Indicators:

Phenolphthalein - 0.1 percent in ethyl alcohol.

Crystal violet - 0.1 percent in glacial acetic acid.

1/ Standardization of perchloric acid - Weigh 0.80 (g) of potassium acid phthalate into a 250 ml beaker. Add 50 ml of glacial acetic acid, warm to hasten solution, cool and wash sides of beaker with glacial acetic acid. Add 3 drops of crystal violet indicator and titrate with the perchloric acid to a blue-green endpoint. Calculate the normality by the equation:

$$N = \frac{\text{grams of potassium acid phthalate}}{0.204 \times \text{ml of perchloric acid}}$$

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4.4.3.2 Assay for diazinon content. Weigh 2.0 grams (g) of the concentrate into a 250 ml separatory funnel; add 125 ml of petroleum ether and 2 drops of phenolphthalein indicator. Add 20 ml of the wash solution and shake the funnel vigorously for 1 minute. The water layer should be basic; if not, add 0.1N sodium hydroxide dropwise with shaking until a persistent pink is obtained. If an emulsion forms and phase separation is delayed, add 50 percent ethanol with shaking until a clean-cut phase separation is obtained. Drain the water layer into a second 250 ml separatory funnel and extract with 50 ml of petroleum ether. After a clean phase separation, discard the aqueous layer. Successively wash the two petroleum ether extracts with 30 ml of distilled water and discard washings. If an emulsion forms, add 50 percent ethanol as above. Combine the petroleum ether extracts in a 500 ml standard tapered Erlenmeyer flask and evaporate on a steam bath to a final volume of 10 to 15 ml. (Caution: Do not prolong heating.) Add 25 ml of acetic anhydride; connect flask to a corresponding standard tapered condenser and reflux 2 hours. Wash down condenser with nitromethane and add an additional 40 ml. Cool; add 3 drops of crystal violet indicator and titrate with 0.1N perchloric acid to a blue-green endpoint. Calculate the percent of weight of diazinon in the insecticide at 77°F (25°C) by the equation:

$$\text{Percent diazinon} = \frac{a \times b \times 30.435}{\text{weight of sample}}$$

Where:

a = Normality of perchloric acid.
b = Number of ml of perchloric acid used.

4.4.4 Flashpoint. The flashpoint shall be determine by the tag closed cup tester in accordance with Method 1101 of Fed. Test Method Std. No. 791.

4.4.5 Emulsion stability. Emulsion stability shall be determined as follows: Place 98 ml of hard water 1/ at 80° ± 5°F (26.7 ± 3°C) into a 250 ml beaker and stir at 1000 revolutions per minute (r.p.m.) 2/. Add 2 ml of the concentrate from a pipette and continue stirring for 1 minute. Pour the emulsion into a 100 ml glass-stoppered graduated cylinder; note the time and examine after 1 hour under strong transmitted light for phase formation and for sediment. Record by volume.

1/ The hard water used shall have the composition:

$CaCl_2$	0.3037 g
$M_gCl_2 \cdot 6H_2O$	0.1388 g
Distilled water	Sufficient to make 1 liter of solution

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2/ The stirrer used shall be power driven and shall consist of a glass rod 10 millimeters (mm) in diameter having a 5 mm diameter rod sealed at right angles to make a T-shaped figure. The stirrer, when positioned, shall just clear the beaker bottom so that it can freely rotate.

4.4.6 Staining and residual odor. The test for staining and residual odor shall be conducted as follows: Place 98 ml of distilled water in a 250 ml beaker; add 2 ml of the concentrate and stir. Immerse a 6-inch (152 mm) square of bleached cotton sheeting in the emulsion, wetting well. Remove excess liquid and hang to dry for 24 hours. Examine for staining and residual odor, using a piece of untreated sheeting for comparison.

4.4.7 pH. The pH of the insecticide shall be tested as follows: Place 98 ml of distilled water (freshly boiled and cooled to room temperature) in a glass-stoppered cylinder; add 2 ml of concentrate and mix by vigorous shaking. Immediately determine the pH of the emulsion by any standard method.

4.4.8 Water content. The water content of the concentrate shall be determined by the distillation procedure outlined in method 3001 of Fed. Test Method Std. No. 791.

4.4.9 Compatability with kerosene and diesel oil. Compatability with kerosene and diesel oil shall be determined as follows: With ingredients at 80° ± 5°F (26.7 ± 3°C), add 2 ml of the concentrate to 98 ml of kerosene contained in glass-stoppered cylinder. Mix thoroughly and examine for homogeneity, haze, and sediment. Allow to stand 24 hours and re-examine. Repeat the above with diesel oil as the diluent.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A. Insecticide shall be furnished in 1-gallon (3.8 l) capacity metal pails conforming to type I, class 3 or 4 of PPP-P-704, The pail shall be provided with a flexible spout closure and a closure seal conforming to PPP-P-704. Pails shall be made of tin plate, hot dipped or electrolytic. Interior coating of containers shall not be required. Exterior plan B coating and side seam striping shall be required.

5.1.2 Level C packaging. Insecticide shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four 1-gallon (3.8 l) metal pails packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V3s of PPP-B-636. Each container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.

5.2.2 Level B packing. Four 1-gallon (3.8 l) metal pails packaged as specified in 5.1 shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, type CF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.2.

5.2.3 Level C packing. Insecticide packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking.

5.3.1 Military requirements. In addition to the marking specified in 3.3 and any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129 and in accordance with the labeling requirements of Title 49, Code of Federal Regulations, Hazardous Materials Regulations.

5.3.2 Civil agencies. In addition to the marking specified in 3.3 and any special marking required by the contract or order, shipments shall be marked in accordance with FED-STD-123 and in accordance with the labeling requirements of Title 49, Code of Federal Regulations, Hazardous Materials Regulations.

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6. NOTES

6.1 Intended use. The diazinon-insecticide covered by this specification is intended for use in controlling household, lawn and ornamental insects and for application in food and non-food areas of food handling establishments.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of container required (see 5.1.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Whether additional marking is required (see 5.3).

6.3 Liquid insecticide is purchased by volume, the unit being a US gallon (3.8 l) at 77°F (25°C).

Custodians:

Army - GL
Navy - SH
Air Force - 68

Review activities:

Army - MD
Navy - YD

User activity:

Navy - MC

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS
AGR - AFS
VA - DMS
HEW - FDA, NIH
DOT - SMS

Project No. 6840-0315

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